Repetition Grundlagen Algebra

Füllen Sie die Kästchen (1 Zahl oder 1 Variable), so dass die Gleichungen stimmen.

1. $(-\square c^2)^2 \cdot (-3)^2 = 36c^4$

2. $2x^2 + 5y^3 + \square x^2 - z^3 + x^2 + \square y^3 - 2z^3 = 6x^2 + 2y^3 - \square z^3$

3. $11a^3 - \square a^4 + 13b - (2a)^2 + (a^2)^2 - 2 \cdot 3a - \frac{\square}{3} b = -a(a^3 - 11a^{\square} + 4a + \square)$

4. $3^4 \cdot 3^7 : 3^{\square} = 3^6$

5. $3x^{\square} z^3 y^2 \cdot (-3xz^2 y^{\square}) = \square x^3 y^6 z^{\square}$

6. $(a^2)^2 \cdot (a^2)^{\square} \cdot a^{(2^3)} = a^{20}$

7. $((-b)^6)^{\square} - (b^2)^9 = 0$

8. $(s^4)^2 \cdot s^{\square} = s^{13}$

9. $d^2(1+e) + (e - \square)(e+d) = d^2 e + e^2$

10. $(q+r)^2 - (q - \square)^2 = \square qr$

11. $\dfrac{xy^{\square}z^2}{x^2y^3z^{\square}} : \dfrac{x^2}{y^2z^3} = x^{\square}y^6z$

12. $\left(\dfrac{f}{g}\right)^{13} : \left(-\dfrac{f}{g}\right)^{8} \cdot (fg)^{\square} = f^3g^{\square}$

13. $(5a^{\square}b^2c^{-3})^{\square} = \frac{1}{25}a^2b^{\square^{\square}}$

14. $(u-\square)^{\square} = \square^5 - 5u^4v + \square u^3v^{\square} + \square u^2v^3 + 5uv^4 + \square v^5$

15. $(\square\square - 3y)^2 = 25x^2 - \square\square\square + 9y^2$

16. $(\square e + \square)^{\square} = 8e^3 + \square e^2f + \square ef^2 + \square^{\square}$

17. $2(3a - \square)(\square + 4b) = 6a^2 + \square ab + \square b^2$

18. $x(2z+3y)(2z - \square\square) = 4\square z^2 - 9xy^2$

* 19. $\left(\dfrac{24a^5}{15x^6} : \dfrac{48a^3}{45x^{\square}}\right) \cdot \left(\dfrac{24a^1s^{\square}}{13sx^2} : \dfrac{8}{26a^{\square}x^7}\right) = \square a^7x^{-11}$

* 20. $\left(\dfrac{a^n \cdot b^3}{a \cdot b^{\square}}\right)^{-1} \cdot \left(\dfrac{a^{2n} \cdot b^4}{a^3 \cdot a^{\square} \cdot b^n}\right)^2 = a^{3n-3}b^{5-n}$